

# Projected Subgradient Optimization with Polyak Steps and Langevin Perturbations for Constrained Single-Cell Representation Learning under Non-Smooth Potentials

## Master Thesis Presentation

Dimitrios Spathoulas

**Supervisor:** Prof. Dr. Yves Moreau

**Assistant-supervisor:** Dr. Susan Ghaderi

**Assessor:** Dr. Vladimir Despotovic

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# Outline

- ① Introduction and Motivation
- ② Objectives and Research Questions
- ③ Proposed Method
- ④ Results
- ⑤ Conclusions
- ⑥ Future Work

## Single-cell RNA Sequencing

- ▶ scRNA-seq measures gene expression at the level of individual cells
- ▶ A central task is to identify groups of cells with similar transcriptional profiles
- ▶ Let the count matrix be:

$$A \in \mathbb{N}^{G \times C}$$

where  $C$  is the number of cells and  $G$  is the number of genes.

- ▶ Challenges:
  - high-dimensional
  - sparse
  - noisy
- ▶ One aims to map this to a low-dimensional representation:

$$A \rightarrow Z \in \mathbb{R}^{k \times C} \quad \text{or} \quad Z \in \mathbb{R}_+^{k \times C}, \quad k \ll G$$

## QC & Preprocessing Pipeline

- ▶ **Quality Control (Filtering):**

Remove technical artifacts to isolate healthy, single cells. Empty droplets or doublets based on extreme total counts, and dead cells.

- ▶ **Library Size Normalization:**

Scale all cells to a uniform total count (e.g.,  $10^4$ ). Ensures comparability and prevents cells with artificially high sequencing depth from dominating.

- ▶ **Log-Normalization:**

Stabilize variance and compresses the dynamic range of highly expressed genes to prevent them from dominating.

- ▶ **HVG Selection:**

Discard the baseline genes every cell needs, keeping only the genes that vary significantly relative to their expected levels (mean expression), assuming these define the cell-type differences.

## Representation Learning & Dimensionality Reduction

- ▶ Principal Component Analysis: Maximize variance by projecting the data using the top  $k$  representative eigenvectors,  $V_k$ , extracted from the covariance matrix of  $A$ :

$$\hat{A} = V_k V_k^\top A$$

- ▶ Non-Negative Matrix Factorization: Parts-based representation by restricting the search space to the non-negative orthant (trained with multiplicative update rules):

$$\hat{A} = WH, \quad W, H \geq 0$$

- ▶ Autoencoders: Defined as learning an  $E : \mathbb{R}^n \rightarrow \mathbb{R}^k$  (encoder network), and  $D : \mathbb{R}^k \rightarrow \mathbb{R}^n$  (decoder network) that minimize the overall distortion function:

$$\arg \min_{E,D} \mathbb{E}[\Delta(x, D(E(x)))]$$

## Objectives of this Thesis

- ▶ **Formulate an NMF-like Interpretable Autoencoder:** Design a network structure that yields biologically interpretable weights and latent, necessitating proximal operators.
- ▶ **Enforce Sparsity Priors:** Incorporate  $\ell_1$  penalty to reflect data structure, creating a non-smooth optimization problem where subgradients are theoretically more appropriate than optimizers like Adam.
- ▶ **Benchmark Optimization Step Rules:** Evaluate the convergence stability and performance of four step-size rules.
- ▶ **Explore the Posterior via Langevin Dynamics:** Transition from MAP estimation to stochastic diffusion to explore solutions near that landscape and construct a more robust, averaged latent representation.

## Research Questions

- 1 Which projected subgradient step-size rule gives the most stable and effective optimization?
- 2 Can subgradients be used with Langevin diffusion and preserve sparsity?
- 3 How does the non-negativity boundary affect diffusion?
- 4 Does stochastic exploration improve the deterministic latent representation under these conditions?
- 5 Can projected subgradients and langevin dynamics be used for constrained single-cell representation learning under sparsity priors?

## The Constrained Non-Smooth Objective

$$\min_W \mathcal{J}(W) = \underbrace{\mathcal{L}(A, \hat{A})}_{\text{reconstruction}} + \underbrace{\lambda_{\text{graph}} \text{Tr}(ZLZ^\top)}_{\text{local structure}} + \underbrace{\lambda_1 \|W\|_1}_{\text{sparsity}} + \underbrace{\lambda_2 \|W\|_F^2}_{\text{weight decay}}$$

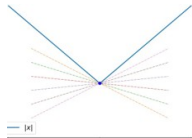
$$\text{s.t. } A, \hat{A} \geq 0, \quad W, Z \geq 0, \quad \hat{A} = WW^\top A, \quad L = D - Q$$

- **Reconstruction:** Main reconstruction loss (Frobenius Norm or KL-Divergence).
- $\ell_1$  **Prior:** Forces sparsity but introduces non-differentiability at zero.
- $\ell_2$  **Prior:** Smoothing-Generalization.
- **Graph Regularization:** Preserve local manifold topology (pulling force).
- **Constrained:** Restricts the search space to the non-negative orthant.

## Non-smoothness, Subgradients and Subdifferentials

- ▶ A function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is non-smooth if  $\nabla f(x)$  is undefined at certain points in its domain:

$$f(x) = \|x\|_1 \implies \nabla f(0) \text{ is undefined.}$$



There is no unique gradient at these non-differentiable points

- ▶ For a convex function, the subdifferential is the closed, convex set of vectors  $g$  satisfying the global linear lower-bound condition:

$$\partial f(x) = \{g \in \mathbb{R}^n \mid f(z) \geq f(x) + \langle g, z - x \rangle, \forall z \in \text{dom } f\}$$

- ▶ At a non-differentiable point, the subdifferential contains multiple valid subgradients.  $-g$  is not necessarily a descent direction (lower bound).

## Subgradient Method & Step Size Strategies

### ► Subgradient Method:

$$x^{k+1} = x^k - \alpha_k g^k \quad \text{selecting: } g^k = \arg \min_{g \in \partial f(x^k)} \|g\|_2$$

### ► Four step rules:

- Constant ( $\alpha_k = \alpha_0$ ): Oscillates around the minimum without converging.
- Diminishing ( $\alpha_k = \alpha_0 / (k + b)^p$ ): Guarantees asymptotic convergence but requires extensive tuning.
- Decaying ( $\alpha_k = \alpha_0 \gamma^k$ ): Unless the initial point is close to optimum or  $\alpha_0$  is large enough, it freezes at a suboptimal point.
- Scaled Polyak: Adaptive method that calculates the step size using the current objective value and an estimate of the optimal loss:

$$\alpha_k = \alpha_0 \frac{f(x^k) - f^*}{c_{po} \|g^k\|_2^2}$$

## Projected Subgradient Method

- ▶ The Euclidean projection of a point  $x \in \mathbb{R}^n$  onto a closed convex set  $C$  is:

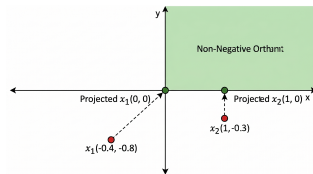
$$\Pi_C(x) = \arg \min_{u \in C} \|u - x\|_2^2$$

- ▶ The operator is non-expansive. For all  $x, y$ :

$$\|\Pi_C(x) - \Pi_C(y)\|_2 \leq \|x - y\|_2$$

- ▶ We take a standard subgradient step, then project the result back into the feasible region:

$$x^{k+1} = \Pi_C(x^k - \alpha_k g^k)$$



## Decomposition of the Objective Function

- ▶ The objective is decomposed into a differentiable non-convex part and a convex non-smooth  $\ell_1$  part:

$$\mathcal{J}(W) = \underbrace{\mathcal{L}(A, \hat{A}) + \lambda_{\text{graph}} \mathcal{L}_{\text{graph}} + \lambda_2 \|W\|_F^2}_{h(W)} + \underbrace{\lambda_1 \|W\|_1}_{c(W)}$$

- ▶ The generalized subdifferential of the full objective is given by the gradient of the smooth part plus the subdifferential of the nonsmooth part:

$$\partial \mathcal{J}(W) = \nabla h(W) + \partial c(W)$$

## Decomposition of the Objective Function (Cont.)

- ▶ The generalized subdifferential  $G^{(k)} \in \partial \mathcal{J}(W^{(k)})$  at iteration  $k$  is computed as:

$$G^{(k)} = \underbrace{\nabla_W \mathcal{L}(A, \hat{A}) + \lambda_{\text{graph}} \nabla_W \mathcal{L}_{\text{graph}} + 2\lambda_2 W^{(k)}}_{\nabla h(W^{(k)})} + \underbrace{\lambda_1 \text{sgn}(W^{(k)})}_{\in \partial c(W^{(k)})}$$

- ▶ The signum function is applied element-wise to each weight  $W_{ij}^{(k)}$ :

$$\text{sgn}(W_{ij}^{(k)}) = \begin{cases} 1 & \text{if } W_{ij}^{(k)} > 0 \\ 0 & \text{if } W_{ij}^{(k)} = 0 \\ -1 & \text{if } W_{ij}^{(k)} < 0 \end{cases}$$

- ▶ The final projected subgradient update rule becomes:

$$W^{(k+1)} = \max(0, W^{(k)} - \alpha_k G^{(k)})$$

## From MAP Estimation to Posterior Sampling

- ▶ MAP Estimate: So far, we have sought the parameter set  $\theta$  that maximizes the posterior probability, which is equivalent to minimizing the potential function  $f(\theta)$ :

$$\theta_{\text{MAP}} = \arg \max_{\theta} \left( p(\mathcal{D} \mid \theta) p(\theta) \right) = \arg \min_{\theta} \underbrace{\left( -\log p(\mathcal{D} \mid \theta) - \log p(\theta) \right)}_{f(\theta)}$$

- ▶ The potential function is non-convex, standard optimization only yields a local minimum, a single point estimate ignores uncertainty.
- ▶ Generating samples from the posterior distribution allows one to capture a robust ensemble of neighboring, high-performing models rather than relying on a single, isolated parameter state.
- ▶ We now aim to sample from the target posterior  $\pi(\theta) \propto e^{-f(\theta)}$ .

## Unadjusted Langevin Algorithm

- ▶ The Unadjusted Langevin Algorithm constructs the MCMC chain using the gradient of the potential:

$$X_{k+1} = X_k - \eta \nabla f(X_k) + \sqrt{2\eta} \xi_k$$

where  $\eta > 0$  is the step size and  $\xi_k \sim \mathcal{N}(0, I)$ .

- ▶ The update combines a deterministic gradient step (drift):

$$-\eta \nabla f(X_k)$$

which moves toward regions of lower potential, and a stochastic diffusion (random perturbation):

$$\sqrt{2\eta} \xi_k$$

which explores the parameter space.

## Projected Subgradient Langevin

- The update then becomes:

$$W^{(k+1)} = \Pi_{\geq 0} \left( W^{(k)} - \alpha_k G^{(k)} + \sqrt{2\alpha_k} \xi^{(k)} \right)$$

Where:

- $G^{(k)} \in \partial \mathcal{J}(W^{(k)})$  is the minimum norm generalized subgradient.
  - $\alpha_k$  the step size at  $k$ .
  - $\xi^{(k)}$  is a matrix with each element drawn i.i.d. from  $\mathcal{N}(0, 1)$ .
- We maintain an average of the network weights across the sampled epochs. At each sampled step  $s$ , the averaged weights are updated incrementally:

$$W_{Lang}^{(s)} = \frac{s-1}{s} W_{Lang}^{(s-1)} + \frac{1}{s} W^{(k_s)}$$

$k_s$  is the epoch at which the  $s$ -th sample is collected.

## Framework

- ▶ **Data Preprocessing:** QC, scaling, log-normalization, HVG selection.
- ▶ **Initialization:** Initialize the non-negative weight matrices.
- ▶ **Objective Formulation:** Define the loss function and hyperparameter weights for the penalties.
- ▶ **MAP Estimation:** Train the model using the projected subgradient method for a predefined number of epochs to find a mode (Point Estimate).
- ▶ **Langevin Phase:** Inject stochastic perturbations starting from the best MAP state, switching to a constant step size rule.
- ▶ **Posterior Sampling and Merging:** Start collecting weight samples once the Markov chain reaches a stationary distribution.
- ▶ **Evaluation:** Assess the model's performance and quality of the representations.

## Evaluation Metrics: Optimization Diagnostics

### ► RMSE:

- Track the objective function value and reconstruction error over training epochs.
- Confirms the convergence of the MAP estimate and the stability/variance of the Langevin exploration.

### ► Sparsity:

- The fraction of parameters (weights and latent values) that fall below a  $10^{-5}$  threshold.
- Evaluates how effectively the  $\ell_1$  penalty and non-negativity constraint collapse redundant parameters.

### ► Weight Norms:

- The Frobenius norm (un-squared  $\ell_2$  magnitude) of the weight and latent.
- Tracks model complexity and highlights norm inflation during noise injection.

## Evaluation Metrics: Quantitative Clustering

- ▶ **External Metrics** (*Compares Predicted Clusters vs. True Labels*):
  - **Adjusted Rand Index (ARI) [-1 to 1]**: Measures how well the predicted clusters group the true cell types together, while penalizing for lucky random guesses. 0 means random guessing.
  - **Normalized Mutual Information (NMI) [0 to 1]**: Measures the "purity" of the clusters. It evaluates how much knowing a cell's assigned cluster reduces the uncertainty about its true biological label.
- ▶ **Internal Metrics** (*Compares Distances in the Latent Space*):
  - **Silhouette Score [-1 to 1]**: Measures how compact a set of cells is within its own cluster, compared to its distance from the nearest neighboring cluster in the latent space.

True Labels = Ground-truths = expert-annotated biological cell types

## Datasets and Baselines

| Dataset                         | Before QC<br>(Cells / Genes) | After QC<br>(Cells / Genes) | After HVGs<br>(Genes) | Latent<br>Dim. |
|---------------------------------|------------------------------|-----------------------------|-----------------------|----------------|
| 10x PBMC 3k (10x Genomics)      | 2,700 / 32,738               | 2,637 / 13,714              | 1,834                 | 15             |
| P14 Mouse Retina (Macosko)      | 49,300 / 24,658              | 31,042 / 20,605             | 2,651                 | 15             |
| Mouse Retinal Bipolar (Shekhar) | 44,994 / 24,904              | 42,767 / 20,557             | 2,483                 | 15             |

### Comparative Baselines:

- ▶ Principal Component Analysis
- ▶ Clamped PCA
- ▶ Non-Negative Matrix Factorization
- ▶ Adam-optimized Autoencoder (with appropriate step-size)

## PBMC: Baselines and Step Rule Comparisons

Baseline Models

| Method      | c  | Sil   | ARI   | NMI   |
|-------------|----|-------|-------|-------|
| PCA         | 9  | 0.185 | 0.638 | 0.785 |
| Clipped PCA | 11 | 0.132 | 0.448 | 0.705 |
| NMF         | 11 | 0.252 | 0.534 | 0.746 |
| Adam        | 10 | 0.226 | 0.473 | 0.697 |

Main Models Results

| Step Rule      | c  | Sil   | ARI   | NMI   |
|----------------|----|-------|-------|-------|
| Constant       | 11 | 0.248 | 0.418 | 0.653 |
| Diminishing    | 11 | 0.253 | 0.411 | 0.650 |
| Decaying       | 10 | 0.252 | 0.446 | 0.679 |
| Scaled Polyak* | 10 | 0.223 | 0.492 | 0.681 |
| Scaled Polyak  | 8  | 0.219 | 0.593 | 0.743 |
| Langevin       | 9  | 0.201 | 0.574 | 0.736 |

*NNDSVD initialization, appropriate step size for all, same regularization hyperparameters,  $\lambda_{graph} = 0$ .*

*\* = Random Initialization*

## PBMC: Loss Trajectories

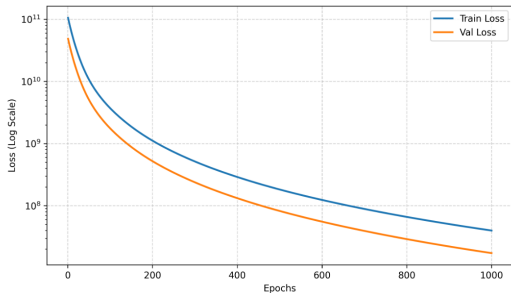


Figure: Loss trajectory under a decaying step-size schedule

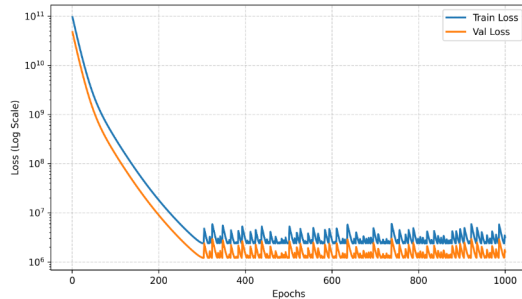


Figure: Loss trajectory under the scaled Polyak step rule.

## PBMC: Sparsity

Sparsity (Threshold:  $10^{-5}$ )

| Method        | Latent | Projector |
|---------------|--------|-----------|
| Langevin      | 0.0000 | 0.0322    |
| Scaled Polyak | 0.0011 | 0.7913    |
| Adam          | 0.0526 | 0.9146    |
| PCA           | 0.0000 | 0.0056    |
| Clipped PCA   | 0.5281 | 0.5433    |
| NMF           | 0.3922 | 0.4198    |

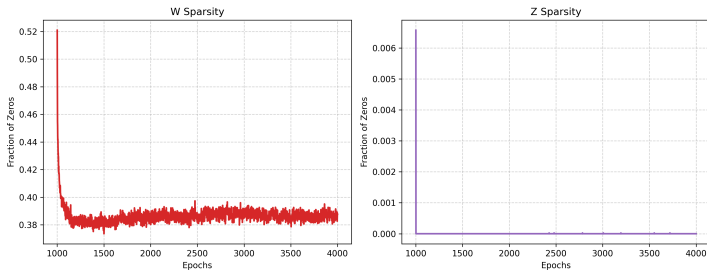


Figure: Sparsity across epochs during Langevin dynamics.

## Langevin Dynamics Across Step Rules

Impact of Langevin Phase Across Step Rules

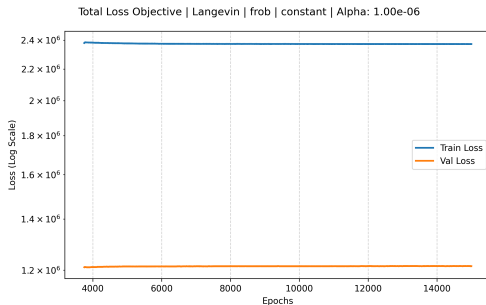
| Step Rule          | Phase         | c        | Sil          | ARI          | NMI          |
|--------------------|---------------|----------|--------------|--------------|--------------|
| <b>Constant</b>    | Deterministic | 11       | 0.248        | 0.418        | 0.653        |
|                    | + Langevin    | 10       | <b>0.286</b> | <b>0.489</b> | <b>0.686</b> |
| <b>Diminishing</b> | Deterministic | 11       | <b>0.253</b> | 0.411        | 0.650        |
|                    | + Langevin    | 11       | 0.240        | <b>0.486</b> | <b>0.687</b> |
| <b>Polyak</b>      | Deterministic | <b>8</b> | <b>0.219</b> | <b>0.593</b> | <b>0.743</b> |
|                    | + Langevin    | 9        | 0.201        | 0.574        | 0.736        |

Langevin diffusion provides corrective boosts to suboptimal local minima, but struggles to exceed the capacity ceiling of the well-converged Polyak solution.

# PBMC: Extended Training and Stacked AE Results

## Impact of Extended Training and SAE on Clustering Metrics

| Model             | c  | Sil   | ARI   | NMI   |
|-------------------|----|-------|-------|-------|
| Polyak            | 8  | 0.219 | 0.593 | 0.743 |
| Langevin          | 9  | 0.201 | 0.574 | 0.736 |
| Lang. Extended    | 10 | 0.229 | 0.561 | 0.731 |
| Stacked AE (Lang) | 8  | 0.221 | 0.602 | 0.746 |



Total loss objective over the extended run. The trajectory spans the post-optimization phase (from epoch 3,750 onward), with posterior samples collected from epoch 10,000.

## P14 Mouse Retina and Mouse Retinal Bipolar Results

P14 Mouse Retina

| Method        | c  | Sil   | ARI   | NMI   |
|---------------|----|-------|-------|-------|
| Langevin      | 19 | 0.255 | 0.341 | 0.591 |
| Scaled Polyak | 18 | 0.244 | 0.321 | 0.596 |
| Adam          | 21 | 0.226 | 0.296 | 0.569 |
| PCA           | 22 | 0.214 | 0.311 | 0.618 |
| PCA Clamped   | 24 | 0.188 | 0.234 | 0.570 |
| NMF           | 25 | 0.221 | 0.206 | 0.580 |

Mouse Retinal Bipolar

| Method        | c  | Sil   | ARI   | NMI   |
|---------------|----|-------|-------|-------|
| Langevin      | 17 | 0.207 | 0.207 | 0.540 |
| Scaled Polyak | 19 | 0.201 | 0.222 | 0.538 |
| Adam          | 19 | 0.169 | 0.219 | 0.535 |
| PCA           | 19 | 0.241 | 0.256 | 0.568 |
| PCA Clamped   | 20 | 0.267 | 0.266 | 0.568 |
| NMF           | 19 | 0.262 | 0.202 | 0.531 |

## P14 Mouse Retina and Mouse Retinal Bipolar Trajectories comparisons (RMSE)

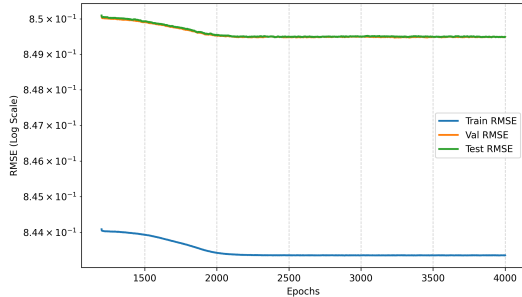


Figure: RMSE trajectory for P14 Mouse Retina.

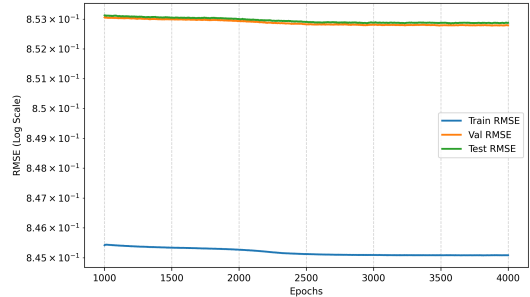


Figure: RMSE trajectory for Mouse Retinal Bipolar.

## P14 Mouse Retina and Mouse Retinal Bipolar Trajectories comparisons (Sparsity)

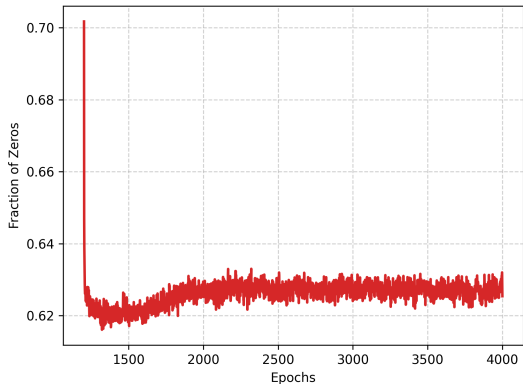


Figure: Sparsity across epochs for P14 Mouse Retina.

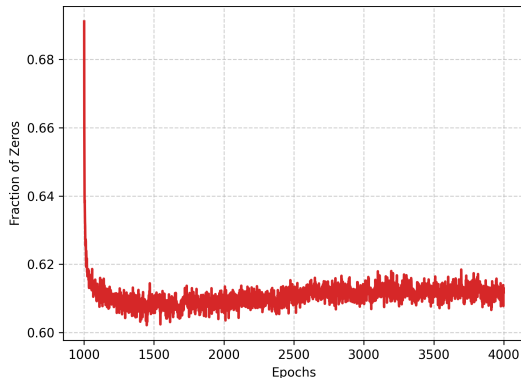


Figure: Sparsity across epochs for Mouse Retinal Bipolar.

## P14 Mouse Retina and Mouse Retinal Bipolar Trajectories comparisons (Weight Norms)

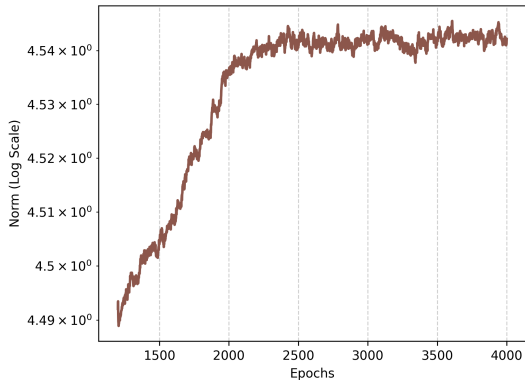


Figure: Weight Norms across epochs for P14 Mouse Retina.

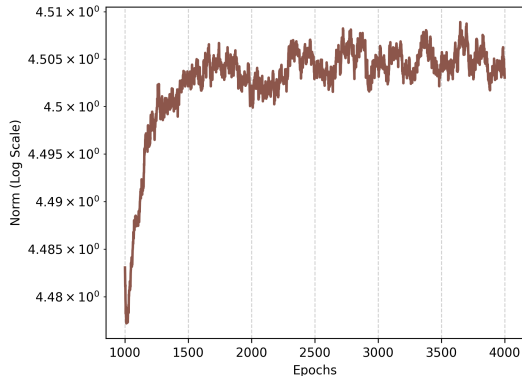


Figure: Weight Norms across epochs for Mouse Retinal Bipolar.

## Conclusions

- ▶ **Initialization is Critical:** Informed initialization is essential. Random weights initialization fails to converge to solutions as good as those with NNDSVD variants.
- ▶ **Superiority of the Polyak Step:** The scaled Polyak rule proved to be the most stable and performant optimization scheme, requiring minimal manual tuning.
- ▶ **Corrective Effect on Weaker Rules:** When paired with sub-optimal step rules, adding Langevin diffusion yields measurable improvements in the final representations.
- ▶ **Representational Capacity Ceiling:** The strict linearity of the architecture imposes a hard limit on its representational capacity. Even when stacked hierarchically, the model cannot be pushed beyond this deterministic ceiling.

## Conclusions (Cont.)

- ▶ **Limited Stochastic Gains:** Unadjusted Langevin dynamics fails to consistently improve upon the well-converged deterministic Polyak solutions across datasets.
- ▶ **Sparsity vs. Diffusion Conflict:** Ideally, Langevin dynamics would explore alternative sparse representations. Continuous isotropic noise constantly fights the sparsity-inducing gradients, perturbing weights away from zero and preventing exact structural sparsity.
- ▶ **Boundary Rectification Bias:** Hard non-negativity projections block negative noise but retain positive perturbations, creating an upward norm drift that distorts the true posterior.

## Future Work

- ▶ **Proximal & Adaptive Techniques:** Transition from subgradients to proximal operators (e.g., soft-thresholding) and adaptive step sizes without requiring an estimate of the optimal function value.
- ▶ **Architectural Expressivity:** Introduce bounded or smooth nonlinear activation functions that compound representational depth without causing gradient explosion under non-negativity constraints. Additionally, relaxing the tied-weight constraint would remove an implicit regularization bottleneck.
- ▶ **Structure-Preserving Regularization:** Incorporate richer graph-based or contrastive regularizers (pushing-pulling forces) beyond the simple graph Laplacian to better maintain the local manifold geometry of the data.

## Future Work (Cont.)

- ▶ **Boundary-Aware Sampling:** Integrate sampling techniques explicitly designed for constrained spaces, such as Mirrored Langevin Dynamics, to properly explore the posterior without biased hard projections.
- ▶ **Metropolis-Hastings Adjustments:** Implement Metropolis-Adjusted Langevin Algorithm or Hamiltonian Monte Carlo to correct the asymptotic bias of unadjusted Langevin diffusion. Will still require careful tuning to mitigate the reduced acceptance rates inherent to high-dimensional state spaces.
- ▶ **Scalability & Calibration:** Leverage greater computational resources to conduct an exhaustive calibration of Langevin diffusion hyperparameters, extending the stochastic evaluation to other single-cell datasets.

# Thank You

**Advanced Master of Science in Artificial Intelligence**

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ESAT-STADIUS-BIOI

*Author:* Dimitrios Spathoulas

*Supervision:* Prof. Yves Moreau

*Assistant-supervisor:* Dr. Susan Ghaderi

*Assessor:* Dr. Vladimir Despotovic

*Questions and Discussion*